

Artificial General Intelligence (AGI): One-Page Summary

Overview

AGI refers to AI systems capable of human-level intelligence across multiple domains, unlike narrow AI which is task-specific. It represents a major shift toward adaptable, general-purpose machine cognition.

Key Concepts

- Narrow AI: Task-specific intelligence (e.g., recommendation systems)
- AGI: Human-like reasoning across domains
- ASI: Intelligence surpassing humans in all aspects

Technical Foundations

- Symbolic AI: Rule-based reasoning systems
- Connectionist AI: Neural networks and deep learning
- Hybrid Models: Combine symbolic + neural approaches (most promising)

Applications

- Finance: Trading, risk management, fraud detection
- Research: Drug discovery, climate modeling
- Business: Automation, decision support, hiring systems

Market Insights

- Market Size: \$2.74B (2023) → \$52B (2032)
- CAGR: ~36.9%
- Finance leads adoption (~38% expected by 2028)

Challenges

- Lack of standard evaluation benchmarks
- High computational & environmental cost
- Ethical concerns: bias, safety, job displacement
- Alignment problem: ensuring AI follows human values

Future Outlook

- Short-term: Hybrid AI systems with limited general intelligence
- Mid-term: Early AGI prototypes in finance & healthcare
- Long-term: Possible ASI emergence and global regulatory frameworks

Key Takeaways

- AGI is both a disruptor and enabler across industries
- Financial sector is the earliest adopter
- Ethical governance and workforce reskilling are critical
- Hybrid architectures are the most viable path forward